

Kuldeep Lodhi

SENIOR FRONTEND ENGINEER

React • TypeScript • Real-Time Systems

Mumbai, India • **Open to relocation: Bengaluru / Hyderabad**

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kuldeep921997.github.io/Portfolio

Summary

Frontend-focused engineer with 7 years building data-intensive enterprise platforms across retail, logistics and fintech. Currently SDE-2 at Reliance Jio, owning the frontend architecture of an inventory platform that serves 12,000+ daily users across 1,900+ stores and 6 retail brands. Strongest in React and TypeScript at scale, real-time interfaces over Kafka and SSE, rendering performance, and design-system work. Leads a 4-engineer frontend team, interviews frontend candidates for the org, and works across the API boundary — REST contract design, plus PostgreSQL query and stored-procedure tuning.

Professional Experience

Reliance Jio Platforms

Jun 2024 – Present

Software Development Engineer II (SDE-2)

Navi Mumbai, India

Software Developer, via Mobile Programming (deputed to Jio)

Dec 2023 – Jun 2024

Continuous tenure on the same platform; converted to Jio payroll as SDE-2 after six months.

- Architected the frontend platform behind Jio's enterprise inventory system — the Store, Cluster and Self-Checkout portals through which store teams manage INR 1,000 Cr of live inventory — working directly with store-operations and brand stakeholders across 6 retail brands
- Established the React and TypeScript foundation the product still runs on, then scaled it into a shared library of 50+ components and a standardized Redux data layer, removing 30% of duplicated UI code and cutting new-module scaffolding from 5 days to 2
- Re-engineered the heaviest inventory dashboards with list virtualization, memoized selectors and route-level code-splitting, cutting time-to-interactive from 8s to 3s on views of 50,000+ rows
- Built the platform's real-time layer, consuming Kafka-published Stock-on-Hand (SOH) and RFID tracking events so store teams see stock movement in under 5 seconds instead of waiting on a 15-minute batch refresh
- Authored the REST contracts and PostgreSQL stored procedures behind high-volume SOH aggregation, cutting the slowest store-level report from 10s to 2s and reducing repeated integration rework between frontend and backend
- Led a 4-engineer frontend team through 30 code reviews per sprint, setting the TypeScript, UI/UX and testing standards the team works to (Jest, React Testing Library)
- Mentored and onboarded 7 engineers onto the platform team, and interviewed 20+ frontend candidates across the org's hiring loop
- Owned the Docker and Azure DevOps CI/CD pipeline that took the team from manual releases to 5–8 automated production deployments per week

Algonauts Advisory

Nov 2021 – Dec 2023

Senior Software Developer

Mumbai, India

- Built the firm's algorithmic trading simulator front to back as its founding engineer — Svelte frontend, Node.js order-execution and portfolio APIs, and market-data ingestion — taking it from prototype to the company's core commercial product
- Designed the data model behind order execution, position keeping and intraday P&L, working directly with the founders to turn trading workflows into schemas that held up as instrument coverage and trade volume grew
- Shipped an ML-driven portfolio recommendation module in Python that lifted user engagement by 25%
- Grew the engineering team from 1 to 3, introducing code review, coding standards and onboarding documentation

Ugam Solutions

Jun 2019 – Nov 2021

Senior Analyst — Data Engineering & Visualization

Thane, India

- Delivered 15 enterprise analytics products for CDC, University of Chicago, Google Screenwise and Ipsos
- Built reusable JavaScript dashboard libraries and SQL stored procedures for survey-data transformation, cutting report generation time by 40% and stabilizing recurring client deliverables

Selected Projects

AnyTrac — RFID Inventory Tracking Platform

React, TypeScript, Redux, Node.js, Kafka, Google Maps SDK

- Shipped the store-facing RFID scanning experience that replaced 5-minute manual stock counts with 50-item carton scans in under a second
- Drove the event-driven design publishing Stock-on-Hand and tracking updates across 15 Kafka topics, letting new analytics consumers subscribe without any change to ingestion
- Built live GPS tracking, route visualization and floor-plan mapping on the Google Maps SDK, giving operations zone-level stock visibility and in-transit exception handling

Impact: Inventory accuracy raised from 75% to 98% and shrinkage reduced by 40%; platform deployed across 1,900+ retail locations

JVA — Retail Video Analytics Platform

React, TypeScript, Konva, Kafka, SSE, NVIDIA Video Analytics

- Built the dashboard and configuration layer over **NVIDIA video-analytics pipelines**, covering footfall, queue monitoring, demographics and heatmap modules across 100+ live camera streams in a 10-store pilot
- Delivered the **Region-of-Interest (ROI) configuration tooling** that lets operations teams define and tune detection zones per camera, so analytics changes no longer require an engineering release
- Implemented Server-Sent Events alerting for fire, intrusion and queue-threshold events, surfacing incidents to store teams within seconds of detection
- Shipped a Konva canvas store planner for floor-plan mapping and camera placement, driving zone-level visitor analytics

Reliance Foundation Scholarship Portal

React, Node.js, MongoDB, OCR / AI Document Parsing

- Migrated a licensed K2 workflow system to a MERN stack, removing per-seat licensing cost and unblocking in-house feature development
- Scaled the platform to 50,000 annual applications on an **AI-backed document-parsing pipeline** — OCR with automated field extraction and validation — removing manual document verification as the process bottleneck

Technical Skills

Languages: TypeScript, JavaScript (ES6+), SQL, Python, HTML5, CSS3

Frontend: React, Redux / Redux Toolkit, Next.js, Svelte, Konva, Tailwind CSS, Material-UI, Chart.js, ApexCharts

Frontend Architecture: Design systems and shared component libraries, state management design, rendering performance (virtualization, memoization, code-splitting), bundle optimization, REST contract design

Real-Time & Streaming: Apache Kafka, Server-Sent Events, event-driven UI patterns, Google Maps SDK

AI Platform Integration: NVIDIA video-analytics pipeline integration, ROI configuration, OCR / AI document-parsing pipelines, ML recommendation systems

Backend & Data: Node.js, Express, PostgreSQL (query optimization, stored procedures), MongoDB, Redis, MySQL

Testing & Quality: Jest, React Testing Library, component and integration testing, accessibility testing

DevOps & Cloud: Git, Docker, Azure DevOps CI/CD, AWS (S3, EC2), ELK (project experience)

Leadership: Team lead (4 engineers), mentoring and onboarding, interviewing, code review, technical design reviews, Agile / Scrum

Education

Bachelor of Engineering, Computer Engineering

2015 – 2019

Vidyalankar Institute of Technology, Mumbai